

Part 1:

Question 1.

A Database Management System (DBMS) is a software system used to manage the data in a structured format. Traditional file systems allowed users to manage data in isolated files, but it came with challenges such as data redundancy, poor security, lack of multi-user support, no backup. These challenges are tackled by DBMS software systems.

Example:

A banking system which is used to process transactions. If two people are trying to withdraw money from the same account, the traditional file system would likely be overwritten or crash causing discrepancies.

Question 2.

In an ER diagram, the strong entity has its own primary key whereas the weak entity depends on the strong entity's primary key to be identified. The primary key of the weak identity is formed using the combination of its own primary key and the primary of the strong entity it depends on.

Question 3.

It should be modeled as a multivalued attribute. The student may have multiple phone numbers which doesn't allow it to be a simple attribute as it allows one value for each student. The composite attribute has sub attributes under it. As the phone number attribute contains multiple values which will make it a multivalued attribute.

Question 4.

A super key is a set of attributes which uniquely identifies a row in a table whereas candidate key contains the minimal attributes that are required to identify a row in a table.

Candidate keys: roll_no, email

Super keys: (roll_no, name), (email, branch)

Question 5.

The constraint that is violated is Foreign key constraint. It ensures that relationships between tables remain valid and cannot reference non-existent data. The DBMS will reject and abort the insert operation.

Question 6.

A bridge table must be created. Its primary key consists of a composite key made up of the primary keys of the participating entity tables. It can't be represented as a column in one of the participating entity tables as it violates the rule of First Normal Form which requires columns to hold single, atomic values.

Part 2.

Part A:

Entities:

Customer: Customer name, phone number, address (composite attribute) – {door no, street/area, pincode}, order id, price of the order

Order: order id, restaurant name, delivery address, price of the order, order details, customer id, quantity

Restaurant: Restaurant name, order id, order details, price of the order

Menu items: item id, item name, price of the items

Relationships:

- 1.) customer places order – 1: N – A customer can place multiple orders, but a specific order is placed by exactly one customer.
- 2.) Restaurant receives order – 1: N – a restaurant can receive many orders, but each order is associated with exactly one order.
- 3.) Restaurant offers menu items: 1: N – A restaurant offers multiple menu items, and each item belongs to one restaurant.
- 4.) Order contains menu items: M: N – An order can contain many menu items, and a menu item can appear in many different orders.

Part B:

Multivalued attribute: phone number as the customer can have multiple phone numbers.

Composite attribute: Delivery address as the address attribute can be divided into sub attributes such as door no, street/area, pincode.

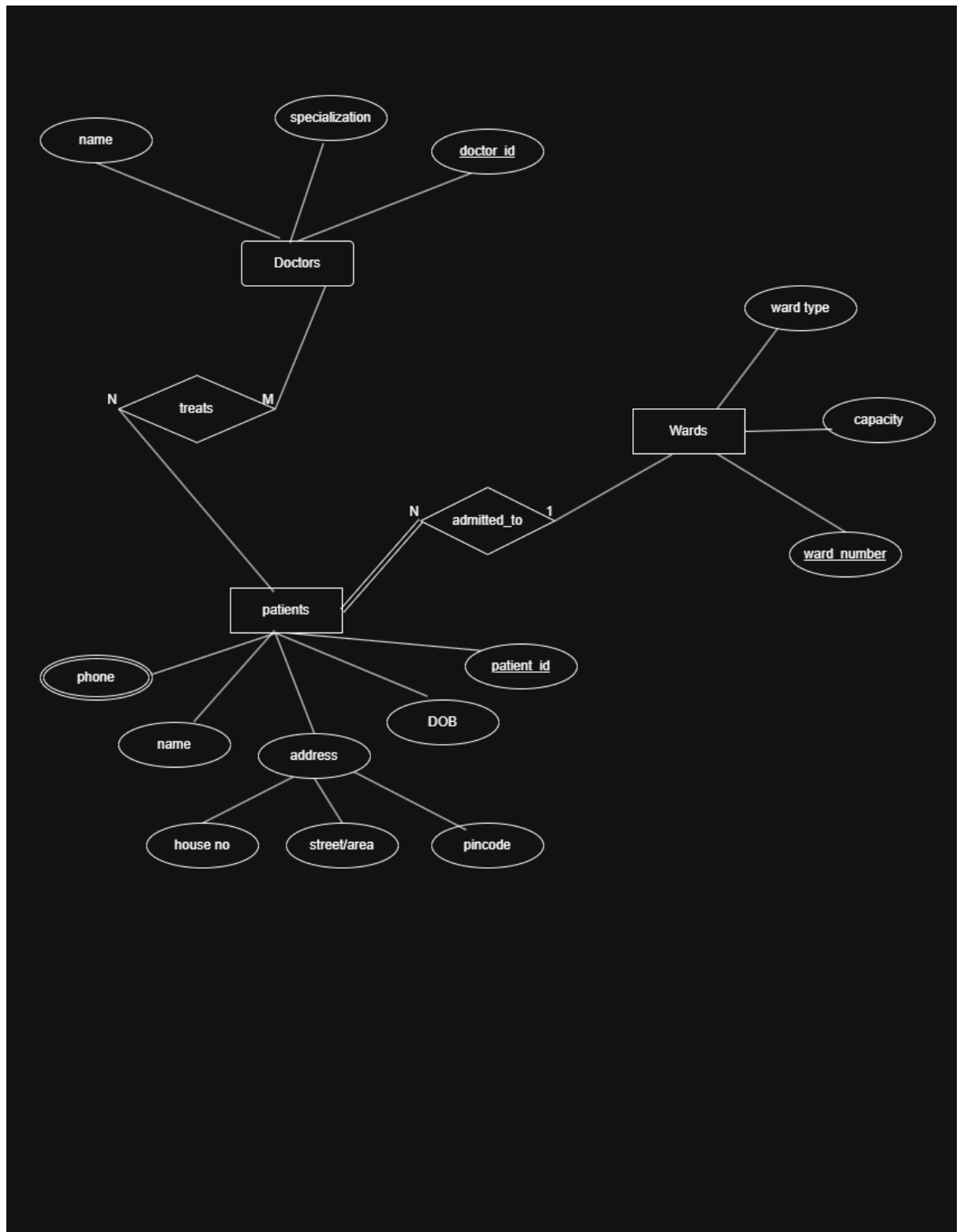
Part C:

OrderItem(order id, item id, quantity)

Primary Key: (order id, item id)

Foreign key: order -> order id. Menu item -> item id

Additional attribute: quantity



Part 3:

Question 2:

A patient can see multiple doctors during their stay and a doctor naturally sees more than one patient. To resolve this in the database schema, Treatment (associative entity) would be created. Its composite primary key is {order_id, patient_id}. The patient entity has a total participation in the admitted_to relationship with the ward entity. A patient cannot exist in the hospital's system without assigned a bed in the ward. The ward entity has partial participation as a new ward might temporarily have zero patients. In the doctor-patient relationship, both entities have partial participation because a doctor may be newly added to the system without treating any patients yet and a patient record may exist before treatment exists. Modeling Address as a composite attribute, instead of storing it in a single field, there are three sub- attributes that it is assigned to – house no, street/area, pincode. Also modeling the phone attribute in the patient entity as multivalued as the patient can give their number and also other numbers of their family/friends in case of emergency.